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Last login: Tue Jun  9 21:02:34 on ttys002
adityadave@Adityas-MacBook-Air-3 ~ % psql -d postgres
psql (18.4)
Type "help" for help.

postgres=# \c tpch
You are now connected to database "tpch" as user "adityadave".
tpch=# -- Step 1: Reset role and disable RLS
RESET ROLE;
ALTER TABLE orders DISABLE ROW LEVEL SECURITY;

-- Step 2: Drop existing view if any
DROP VIEW IF EXISTS orders_secure;

-- Step 3: Create SECURITY BARRIER view with same filter logic as RLS policy
CREATE VIEW orders_secure WITH (security_barrier = true) AS
SELECT o.*
FROM orders o
WHERE EXISTS (
    SELECT 1 FROM customer c
    WHERE c.c_custkey = o.o_custkey
    AND c.c_acctbal > 5000
);

-- Step 4: Grant permissions and set role
GRANT SELECT ON orders_secure TO alice;
GRANT SELECT ON customer TO alice;
GRANT SELECT ON lineitem TO alice;
SET ROLE alice;

-- Step 5: Run Q4 with timing
EXPLAIN (ANALYZE, VERBOSE, BUFFERS)
SELECT
    o_orderpriority,
    COUNT(*) AS order_count
FROM
    orders_secure
WHERE
    o_orderdate >= DATE '1994-01-01'
    AND o_orderdate < DATE '1994-01-01' + INTERVAL '3' MONTH
    AND EXISTS (
        SELECT *
        FROM lineitem
        WHERE l_orderkey = o_orderkey
        AND l_commitdate < l_receiptdate
    )
GROUP BY
    o_orderpriority
ORDER BY
    o_orderpriority;
WARNING: ===== RAW PARSE TREE =====
DETAIL:  {RAWSTMT
:stmt
{VARIABLESETSTMT
:kind 4
:name role
:args <>
:jumble_args false
:is_local false
:location -1
}
:stmt_location 0
:stmt_len 10
}

WARNING: ===== ANALYZED QUERY =====
DETAIL:  {QUERY
:commandType 6
:querySource 0
:canSetTag true
:utilityStmt
{VARIABLESETSTMT
:kind 4
:name role
:args <>
:jumble_args false
:is_local false
:location -1
}
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:cteList <>
:rtable <>
:rteperminfos <>
:jointree <>
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList <>
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDecls <>
:withCheckOptions <>
:stmt_location 0
:stmt_len 10
}

RESET
WARNING: ===== RAW PARSE TREE =====
DETAIL:  {RAWSTMT
:stmt
{ALERTABLESTMT
:relation
{RANGEVAR
:catalogname <>
:schemaname <>
:relname orders
:inh true
:relpersistence p
:alias <>
:location 12
}
}
:cmds (
{ALERTABLECMD
:subtype 55
:name <>
:num 0
:newowner <>
:def <>
:behavior 0
:missing_ok false
:recurse false
}
)

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    )
    :objtype 41
    :missing_ok false
  }
:stmt_location 0
:stmt_len 45
}

WARNING: ===== ANALYZED QUERY =====
DETAIL: {QUERY
:commandType 6
:querySource 0
:canSetTag true
:utilityStmt
{ALERTTABLESTMT
:relation
{RANGEVAR
:catalogname <>
:schemaname <>
:relname orders
:inh true
:relpersistence p
:alias <>
:location 12
}
:cmds (
{ALERTTABLECMD
:subtype 55
:name <>
:num 0
:newowner <>
:def <>
:behavior 0
:missing_ok false
:recurse false
}
)
:objtype 41
:missing_ok false
}
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:cteList <>
:rtable <>
:rteperminfos <>
:jointree <>
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList <>
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location 0
:stmt_len 45
}

ALTER TABLE
WARNING: ===== RAW PARSE TREE =====
DETAIL: {RAWSTMT
:stmt
{DROPSMT
:objects ({("orders_secure"))
:removeType 51
:behavior 0
:missing_ok true
:concurrent false
}
:stmt_location 0
:stmt_len 33
}

WARNING: ===== ANALYZED QUERY =====
DETAIL: {QUERY
:commandType 6
:querySource 0
:canSetTag true
:utilityStmt
{DROPSMT
:objects ({("orders_secure"))
:removeType 51
:behavior 0
:missing_ok true
:concurrent false
}
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:cteList <>
:rtable <>
:rteperminfos <>
:jointree <>
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList <>
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>

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:constraintDeps <>
:withCheckOptions <>
:stmt_location 0
:stmt_len 33
}

DROP VIEW
WARNING: ===== RAW PARSE TREE =====
DETAIL: {RAWTMT
:stmt
{VIEWSTMT
:view
{RANGEVAR
:catalogname <>
:schemaname <>
:relname orders_secure
:inh true
:relpersistence p
:alias <>
:location 12
}
:aliases <>
:query
{SELECTSTMT
:distinctClause <>
:intoClause <>
:targetList (
{RESTART
:name <>
:indirection <>
:val
{COLUMNREF
:fields ("o"
{A_STAR
}
}
):location 67
}
:location 67
}
)
:fromClause (
{RANGEVAR
:catalogname <>
:schemaname <>
:relname orders
:inh true
:relpersistence p
:alias
{ALIAS
:aliasname o
:colnames <>
}
):location 76
}
)
:whereClause
{SUBLINK
:subLinkType 0
:subLinkId 0
:testexpr <>
:operName <>
:subselect
{SELECTSTMT
:distinctClause <>
:intoClause <>
:targetList (
{RESTART
:name <>
:indirection <>
:val
{A_CONST
:val 1
:location 111
}
):location 111
}
)
:fromClause (
{RANGEVAR
:catalogname <>
:schemaname <>
:relname customer
:inh true
:relpersistence p
:alias
{ALIAS
:aliasname c
:colnames <>
}
):location 118
}
)
:whereClause
{BOOLEXP
:boolop and
:args (
{A_EXPR
:name ("=")
:lexpr
{COLUMNREF
:fields ("c" "c_custkey")
:location 139
}
:rexpr
{COLUMNREF
:fields ("o" "o_custkey")
:location 153
}
:rexpr_list_start 0
:rexpr_list_end 0
:location 151
}
{A_EXPR
:name (">")
:lexpr
{COLUMNREF
:fields ("c" "c_acctbal")
:location 173
}
:rexpr
{A_CONST
:val 5000
:location 187
}
:rexpr_list_start 0
:rexpr_list_end 0
:location 185
}
):location 169
}
:groupClause <>
:groupDistinct false
:havingClause <>
>windowClause <>
:valueLists <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:lockingClause <>
:withClause <>
:op 0
:all false
:larg <>
:rarg <>
}
}

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        :location 91
    }
    :groupClause <>
    :groupDistinct false
    :havingClause <>
    :windowClause <>
    :valueLists <>
    :sortClause <>
    :limitOffset <>
    :limitCount <>
    :limitOption 0
    :lockingClause <>
    :withClause <>
    :op 0
    :all false
    :larg <>
    :rarg <>
    }
    :replace false
    :options (
        {DEFELEM
        :defnamespace <>
        :defname security_barrier
        :arg "true"
        :defaction 0
        :location 32
        }
    )
    :withCheckOption 0
    }
    :stmt_location 0
    :stmt_len 193
    }
}

WARNING: ===== ANALYZED QUERY =====
DETAIL: {QUERY
:commandType 6
:querySource 0
:canSetTag true
:utilityStmt
{VIEWSTMT
:view
{RANGEVAR
:catalogname <>
:schemaname <>
:relname orders_secure
:inh true
:relpersistence p
:alias <>
:location 12
}
}
:aliases <>
:query
{SELECTSTMT
:distinctClause <>
:intoClause <>
:targetList (
{RETARGET
:name <>
:indirection <>
:val
{COLUMNREF
:fields ("o"
{A_STAR
}
)
}
:location 67
}
}
:location 67
}
)
:fromClause (
{RANGEVAR
:catalogname <>
:schemaname <>
:relname orders
:inh true
:relpersistence p
:alias
{ALIAS
:aliasname o
:colnames <>
}
}
:location 76
}
)
:whereClause
{SUBLINK
:sublinkType 0
:sublinkId 0
:testexpr <>
:operName <>
:subselect
{SELECTSTMT
:distinctClause <>
:intoClause <>
:targetList (
{RETARGET
:name <>
:indirection <>
:val
{A_CONST
:val 1
:location 111
}
}
:location 111
}
)
:fromClause (
{RANGEVAR
:catalogname <>
:schemaname <>
:relname customer
:inh true
:relpersistence p
:alias
{ALIAS
:aliasname c
:colnames <>
}
}
:location 118
}
)
:whereClause
{BOOLEXP
:boolop and
:args (
{A_EXPR
:name ("=")
:lexpr
{COLUMNREF
:fields ("c" "c_custkey")
:location 139
}
}
:rexpr
{COLUMNREF
:fields ("o" "o_custkey")
:location 153
}
}
:rexpr_list_start 0
:rexpr_list_end 0
:location 151
}
{A_EXPR
:name (">")
:lexpr
{COLUMNREF
:fields ("c" "c_acctbal")

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        :location 173
    }
    :rexpr
    (A_CONST
     :val 5000
     :location 187
    )
    :rexpr_list_start 0
    :rexpr_list_end 0
    :location 185
  )
  :location 169
}
:groupClause <>
:groupDistinct false
:havingClause <>
>windowClause <>
:valuesLists <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:lockingClause <>
:withClause <>
:op 0
:all false
:larg <>
:rarg <>
}
:location 91
}
:groupClause <>
:groupDistinct false
:havingClause <>
>windowClause <>
:valuesLists <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:lockingClause <>
:withClause <>
:op 0
:all false
:larg <>
:rarg <>
}
:replace false
:options (
  {DEFELEM
   :defnamespace <>
   :defname security_barrier
   :arg "true"
   :defaction 0
   :location 32
  }
)
:withCheckOption 0
}
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:cteList <>
:rtable <>
:rteperminfos <>
:jointree <>
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList <>
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location 0
:stmt_len 193
}

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WARNING: ===== ANALYZED QUERY =====

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DETAIL:      {QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks true
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:cteList <>
:rtable (
  {RANGETBENTRY
   :alias
   {ALIAS
    :aliasname o
    :colnames <>
   }
  :eref
  {ALIAS
   :aliasname o
   :colnames ("o_orderkey" "o_custkey" "o_orderstatus" "o_totalprice" "o_orderdate" "o_orderpriority" "o_clerk" "o_shippriority" "o_comment")
  }
  :rtckind 0
  :relid 16561
  :inh true
  :relkind r
  :rellockmode 1
  :perminfoindex 1
  :tablesample <>
  :lateral false
  :inFromCl true
  :securityQuals <>
}
)
:rteperminfos (

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{RTEPERMISSIONINFO
:relid 16561
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 9 10 11 12 13 14 15 16)
:insertedCols (b)
:updatedCols (b)
}
)
:jointree
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 1
}
)
:quals
{SUBLINK
:subLinkType 0
:subLinkId 0
:testexpr <>
:operName <>
:subselect
{QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSublinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:ctelist <>
:rttable (
{RANGETBENTRY
:alias
{ALIAS
:aliasname c
:colnames <>
}
:eref
{ALIAS
:aliasname c
:colnames ("c_custkey" "c_name" "c_address" "c_nationkey" "c
_phone" "c_acctbal" "c_mktsegment" "c_comment")
}
:rtkind 0
:relid 16546
:inh true
:relkind r
:rellockmode 1
:perminfoindex 1
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
)
:rtperminfos (
{RTEPERMISSIONINFO
:relid 16546
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 13)
:insertedCols (b)
:updatedCols (b)
}
)
:jointree
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 1
}
)
:quals
{BOOLEXP
:boolop and
:args (
{OPEXP
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 1
:varstype 23
:varstypepm -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location 139
}
{VAR
:varno 1
:varattno 2
:varstype 23
:varstypepm -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 1
:varreturningtype 0
:varnosyn 1
:varattnosyn 2
:location 153
}
)
:location 151
}
{OPEXP
:opno 674
:opfuncid 297
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 6
:varstype 701
:varstypepm -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 6
:location 173
}
}
{FUNCEXP

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:funcid 316
:funcresulttype 701
:funcresultset false
:funcvariadic false
:funcformat 2
:funccollid 0
:inputcollid 0
:args (
  {CONST
    :consttype 23
    :consttypmod -1
    :constcollid 0
    :constlen 4
    :constbyval true
    :constisnull false
    :location 187
    :constvalue 4 [ -120 19 0 0 0 0 0 0 ]
  }
)
:location -1
}
}
:location 185
}
}
:location 169
}
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
  {TARGETENTRY
    :expr
    {CONST
      :consttype 23
      :consttypmod -1
      :constcollid 0
      :constlen 4
      :constbyval true
      :constisnull false
      :location 111
      :constvalue 4 [ 1 0 0 0 0 0 0 0 ]
    }
    :resno 1
    :resname ?column?
    :ressortgroupref 0
    :resorigtbl 0
    :resorigcol 0
    :resjunk false
  }
)
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location 0
:stmt_len 0
}
:location 91
}
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
  {TARGETENTRY
    :expr
    {VAR
      :varno 1
      :varattno 1
      :vartype 23
      :vartypmod -1
      :varcollid 0
      :varnullingrels (b)
      :varlevelsup 0
      :varreturningtype 0
      :varnosyn 1
      :varattnosyn 1
      :location 67
    }
    :resno 1
    :resname o_orderkey
    :ressortgroupref 0
    :resorigtbl 16561
    :resorigcol 1
    :resjunk false
  }
  {TARGETENTRY
    :expr
    {VAR
      :varno 1
      :varattno 2
      :vartype 23
      :vartypmod -1
      :varcollid 0
      :varnullingrels (b)
      :varlevelsup 0
      :varreturningtype 0
      :varnosyn 1
      :varattnosyn 2
      :location 67
    }
  }
  :resno 2
  :resname o_custkey
  :ressortgroupref 0
  :resorigtbl 16561
  :resorigcol 2
  :resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 3
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 3
:location 67
}
:resno 3
:resname o_orderstatus
:ressortgroupref 0
:resorigtbl 16561
:resorigcol 3
:resjunk false
}
{TARGETENTRY

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:expr
{VAR
:varno 1
:varattno 4
:vartype 701
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 4
:location 67
}
:resno 4
:resname o_totalprice
:ressortgroupref 0
:resorigtbl 16561
:resorigcol 4
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 5
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 5
:location 67
}
:resno 5
:resname o_orderdate
:ressortgroupref 0
:resorigtbl 16561
:resorigcol 5
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 6
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 6
:location 67
}
:resno 6
:resname o_orderpriority
:ressortgroupref 0
:resorigtbl 16561
:resorigcol 6
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 7
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 7
:location 67
}
:resno 7
:resname o_clerk
:ressortgroupref 0
:resorigtbl 16561
:resorigcol 7
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 8
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 8
:location 67
}
:resno 8
:resname o_shippriority
:ressortgroupref 0
:resorigtbl 16561
:resorigcol 8
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 9
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 9
:location 67
}
:resno 9
:resname o_comment
:ressortgroupref 0
:resorigtbl 16561
:resorigcol 9
:resjunk false
}
}
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>

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:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location 0
:stmt_len 193
}

CREATE VIEW
WARNING: ===== RAW PARSE TREE =====
DETAIL: {RAWSTMT
:stmt
{GRANTSTMT
:is_grant true
:targetype 0
:objtype 41
:objects (
{RANGEVAR
:catalogname <>
:schemaname <>
:relname orders_secure
:inh true
:relpersistence p
:alias <>
:location 16
}
)
:privileges (
{ACCESSPRIV
:priv_name select
:cols <>
}
)
:grantees (
{ROLESPEC
:roletype 0
:rolename alice
:location 33
}
)
:grant_option false
:grantor <>
:behavior 0
}
:stmt_location 0
:stmt_len 38
}

```

```

WARNING: ===== ANALYZED QUERY =====
DETAIL: {QUERY
:commandType 6
:querySource 0
:canSetTag true
:utilityStmt
{GRANTSTMT
:is_grant true
:targetype 0
:objtype 41
:objects (
{RANGEVAR
:catalogname <>
:schemaname <>
:relname orders_secure
:inh true
:relpersistence p
:alias <>
:location 16
}
)
:privileges (
{ACCESSPRIV
:priv_name select
:cols <>
}
)
:grantees (
{ROLESPEC
:roletype 0
:rolename alice
:location 33
}
)
:grant_option false
:grantor <>
:behavior 0
}
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSublinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:ctcList <>
:rtable <>
:rteperminfos <>
:jointree <>
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList <>
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location 0
:stmt_len 38
}

```

```

GRANT
WARNING: ===== RAW PARSE TREE =====
DETAIL: {RAWSTMT
:stmt
{GRANTSTMT
:is_grant true
:targetype 0
:objtype 41
:objects (
{RANGEVAR
:catalogname <>
:schemaname <>
:relname customer
:inh true
:relpersistence p
:alias <>
:location 16
}
)
:privileges (

```

```

        {ACCESSPRIV
        :priv_name select
        :cols <>
        }
    }
    :grantees (
    {ROLESPEC
    :roletype 0
    :rolename alice
    :location 28
    }
    )
    :grant_option false
    :grantor <>
    :behavior 0
    }
    :stmt_location 0
    :stmt_len 33
    }
}

```

WARNING: ===== ANALYZED QUERY =====:

```

DETAIL: {QUERY
:commandType 6
:querySource 0
:canSetTag true
:utilityStmt
{GRANTSTMT
:is_grant true
:targetType 0
:objType 41
:objects (
{RANGEVAR
:catalogname <>
:schemaname <>
:reName customer
:inh true
:relPersistence p
:alias <>
:location 16
}
)
:privileges (
{ACCESSPRIV
:priv_name select
:cols <>
}
)
:grantees (
{ROLESPEC
:roletype 0
:rolename alice
:location 28
}
)
:grant_option false
:grantor <>
:behavior 0
}
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSublinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTe false
:isReturn false
:ctelist <>
:rtable <>
:rteperminfos <>
:jointree <>
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList <>
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location 0
:stmt_len 33
}
}

```

GRANT

WARNING: ===== RAW PARSE TREE =====:

```

DETAIL: {RAWSTMT
:stmt
{GRANTSTMT
:is_grant true
:targetType 0
:objType 41
:objects (
{RANGEVAR
:catalogname <>
:schemaname <>
:reName lineitem
:inh true
:relPersistence p
:alias <>
:location 16
}
)
:privileges (
{ACCESSPRIV
:priv_name select
:cols <>
}
)
:grantees (
{ROLESPEC
:roletype 0
:rolename alice
:location 28
}
)
:grant_option false
:grantor <>
:behavior 0
}
:stmt_location 0
:stmt_len 33
}
}

```

WARNING: ===== ANALYZED QUERY =====:

```

DETAIL: {QUERY
:commandType 6
:querySource 0
:canSetTag true
:utilityStmt
{GRANTSTMT

```

```

:is_grant true
:targettype 0
:objtype 41
:objects (
  {RANGEVAR
    :catalogname <>
    :schemaname <>
    :relname lineitem
    :inh true
    :reipersistence p
    :alias <>
    :location 16
  }
)
:privileges (
  {ACCESSPRIV
    :priv_name select
    :cols <>
  }
)
:grantees (
  {ROLESPEC
    :roletype 0
    :rolename alice
    :location 28
  }
)
:grant_option false
:grantor <>
:behavior 0
}

:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:ctelist <>
:rtable <>
:rteperminfos <>
:jointree <>
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList <>
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location 0
:stmt_len 33
}

GRANT
WARNING: ===== RAW PARSE TREE =====
DETAIL: {RAWSTMT
:stmt
  {VARIABLESETSTMT
    :kind 0
    :name role
    :args (
      {A_CONST
        :val "alice"
        :location 9
      }
    )
    :jumble_args false
    :is_local false
    :location 9
  }
:stmt_location 0
:stmt_len 14
}

WARNING: ===== ANALYZED QUERY =====
DETAIL: {QUERY
:commandType 6
:querySource 0
:canSetTag true
:utilityStmt
  {VARIABLESETSTMT
    :kind 0
    :name role
    :args (
      {A_CONST
        :val "alice"
        :location 9
      }
    )
    :jumble_args false
    :is_local false
    :location 9
  }
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:ctelist <>
:rtable <>
:rteperminfos <>
:jointree <>
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList <>
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
}

```

```

:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location 0
:stmt_len 14
}

SET
WARNING: ===== RAW PARSE TREE =====
DETAIL: {RAWSTMT
:stmt
{EXPLAINSTMT
:query
{SELECTSTMT
:distinctClause <>
:intoClause <>
:targetList (
{RESTARTGET
:name <>
:indirection <>
:val
{COLUMNREF
:fields ("o_orderpriority")
:location 47
}
:location 47
}
{RESTARTGET
:name order_count
:indirection <>
:val
{FUNCCALL
:funcname ("count")
:args <>
:agg_order <>
:agg_filter <>
:over <>
:agg_within_group false
:agg_star true
:agg_distinct false
:func_variadic false
:funcformat 0
:location 68
}
:location 68
}
)
:fromClause (
{RANGEVAR
:catalogname <>
:schemaname <>
:relname orders_secure
:inh true
:relpersistence p
:alias <>
:location 101
}
)
:whereClause
{BOOLEXP
:boolop and
:args (
{A_EXPR
:name (">=")
:lexpr
{COLUMNREF
:fields ("o_orderdate")
:location 125
}
:rexpr
{TYPECAST
:arg
{A_CONST
:val "\1994-01-01"
:location 145
}
:typeName
{TYPENAME
:names ("date")
:typeOid 0
:setof false
:pct_type false
:typmods <>
:typemod -1
:arrayBounds <>
:location 140
}
:location -1
}
:rexpr_list_start 0
:rexpr_list_end 0
:location 137
}
{A_EXPR
:name ("<")
:lexpr
{COLUMNREF
:fields ("o_orderdate")
:location 166
}
:rexpr
{A_EXPR
:name ("+")
:lexpr
{TYPECAST
:arg
{A_CONST
:val "\1994-01-01"
:location 185
}
:typeName
{TYPENAME
:names ("date")
:typeOid 0
:setof false
:pct_type false
:typmods <>
:typemod -1
:arrayBounds <>
:location 180
}
:location -1
}
:rexpr
{TYPECAST
:arg
{A_CONST
:val "\3"
:location 209
}
:typeName
{TYPENAME
:names ("pg_catalog" "interval")
:typeOid 0
:setof false
:pct_type false
:typmods (
{A_CONST
:val 2
:location 213
}
)
:typemod -1
:arrayBounds <>
:location 200
}
}
}
}

```

```

        :location -1
    }
    :rexpr_list_start 0
    :rexpr_list_end 0
    :location 198
    }
    :rexpr_list_start 0
    :rexpr_list_end 0
    :location 178
    }
    {SUBLINK
    :subLinkType 0
    :subLinkId 0
    :testexpr <>
    :operName <>
    :subselect
    {SELECTSTMT
    :distinctClause <>
    :intoClause <>
    :targetList (
    {RESTARTARGET
    :name <>
    :indirection <>
    :val
    {COLUMNREF
    :fields (
    {A_STAR
    }
    )
    }
    :location 251
    }
    :location 251
    }
    )
    :fromClause (
    {RANGEVAR
    :catalogname <>
    :schemaname <>
    :relname lineitem
    :inh true
    :relpersistance p
    :alias <>
    :location 266
    }
    )
    :whereClause
    {BOOLEXP
    }
    :boolop and
    :args (
    {A_EXPR
    :name ("=")
    :lexpr
    {COLUMNREF
    :fields ("l_orderkey")
    :location 289
    }
    :rexpr
    {COLUMNREF
    :fields ("o_orderkey")
    :location 302
    }
    }
    :rexpr_list_start 0
    :rexpr_list_end 0
    :location 300
    }
    {A_EXPR
    :name ("<")
    :lexpr
    {COLUMNREF
    :fields ("l_commitdate")
    :location 327
    }
    :rexpr
    {COLUMNREF
    :fields ("l_receiptdate")
    :location 342
    }
    }
    :rexpr_list_start 0
    :rexpr_list_end 0
    :location 340
    }
    )
    :location 323
    }
    :groupClause <>
    :groupDistinct false
    :havingClause <>
    :windowClause <>
    :valuesLists <>
    :sortClause <>
    :limitOffset <>
    :limitCount <>
    :limitOption 0
    :lockingClause <>
    :withClause <>
    :op 0
    :all false
    :larg <>
    :rarg <>
    }
    :location 227
    }
    )
    :location 162
    }
    :groupClause (
    {COLUMNREF
    :fields ("o_orderpriority")
    :location 375
    }
    )
    :groupDistinct false
    :havingClause <>
    :windowClause <>
    :valuesLists <>
    :sortClause (
    {SORTBY
    :node
    {COLUMNREF
    :fields ("o_orderpriority")
    :location 404
    }
    :sortby_dir 0
    :sortby_nulls 0
    :useOp <>
    :location -1
    }
    )
    :limitOffset <>
    :limitCount <>
    :limitOption 0
    :lockingClause <>
    :withClause <>
    :op 0
    :all false
    :larg <>
    :rarg <>
    }
    )
    :options (
    {DEFELEM
    :defnamespace <>
    :defname analyze
    :arg <>
    :defaction 0
    :location 9
    }
    {DEFELEM
    :defnamespace <>

```

```

:define verbose
:arg <>
:defaction 0
:location 18
}
{DEFELEM
:definespace <>
:define buffers
:arg <>
:defaction 0
:location 27
}
}
}
:stmt_location 0
:stmt_len 419
}

WARNING: ===== ANALYZED QUERY =====
DETAIL: {QUERY
:commandType 6
:querySource 0
:canSetTag true
:utilityStmt
{EXPLAINSTMT
:query
{QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs true
:hasWindowFuncs false
:hasTargetSRFs false
:hasSublinks true
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE true
:isReturn false
:otelist <>
:rtable (
{RANGETBLENTTRY
:alias <>
:eref
{ALIAS
:aliasname orders_secure
:colnames ("o_orderkey" "o_custkey" "o_orderstatus" "o_totalpri
ce" "o_orderdate" "o_orderpriority" "o_clerk" "o_shippriority"
"o_comment")
}
:rtekind 0
:relid 16847
:inh true
:relkind v
:rellockmode 1
:perminfoindex 1
:tablesample <>
:lateral false
:inFromCI true
:securityQuals <>
}
{RANGETBLENTTRY
:alias <>
:eref
{ALIAS
:aliasname #GROUP#
:colnames ("o_orderpriority")
}
:rtekind 9
:groupexprs (
{VAR
:varno 1
:varattno 6
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 6
:location 47
}
)
:lateral false
:inFromCI false
:securityQuals <>
}
)
:rteperminfos (
{RTEPERMISSIONINFO
:relid 16847
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 12 13)
:insertedCols (b)
:updatedCols (b)
}
)
:jointree
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 1
}
)
:quals
{BOOLEXP
:boolop and
:args (
{OPEXP
:opno 1098
:opfuncid 1090
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 5
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 5
:location 125
}
{CONST
:consttype 1082
:consttypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location 145
:constvalue 4 [ 113 -9 -1 -1 -1 -1 -1 -1 ]
}
)
}
:location 137
}

```

```

}
{OPEXPR
:opno 2345
:opfuncid 2338
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
  {VAR
:varno 1
:varattno 5
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 5
:location 166
}
  {OPEXPR
:opno 1076
:opfuncid 2071
:opresulttype 1114
:opretset false
:opcollid 0
:inputcollid 0
:args (
  {CONST
:consttype 1082
:consttypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location 185
:constvalue 4 [ 113 -9 -1 -1 -1 -1 -1 -1 ]
}
  {CONST
:consttype 1186
:consttypmod 196607
:constcollid 0
:constlen 16
:constbyval false
:constisnull false
:location 209
:constvalue 16 [ 0 0 0 0 0 0 0 0 0 0 0 3 0 0 0 0 ]
}
)
:location 198
}
)
:location 178
}
{SUBLINK
:subLinkType 0
:subLinkId 0
:testexpr <>
:operName <>
:subselect
  {QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:ctelist <>
:rtable (
  {RANGETBLENTRY
:alias <>
:eref
  {ALIAS
:aliasname lineitem
:colnames ("l_orderkey" "l_partkey" "l_suppkey" "l_linenum" "l_quantity" "l_extendedprice" "l_discount" "l_tax" "l_returnflag" "l_linestatus" "l_shipdate" "l_commitdate" "l_receiptdate" "l_shipinstruct" "l_shipmode" "l_comment")
}
:rtekind 0
:relid 16566
:inh true
:relkind F
:rellockmode 1
:perminfoindex 1
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
)
:rteperminfos (
  {RTEPERMISSIONINFO
:relid 16566
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23)
:insertedCols (b)
:updatedCols (b)
}
)
:jointree
  {FROMEXPR
:fromlist (
  {RANGETBLREF
:rtindex 1
}
)
:quals
  {BOOLEXPR
:boolop and
:args (
  {OPEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
  {VAR
:varno 1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location 289
}
)
}
)
}
}

```

```

        {VAR
        :varno 1
        :varattno 1
        :vartype 23
        :vartypmod -1
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 1
        :varreturningtype 0
        :varnosyn 1
        :varattnosyn 1
        :location 302
        }
    )
    :location 300
}
{OPEXPR
:opno 1095
:opfuncid 1087
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 12
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 12
:location 327
}
{VAR
:varno 1
:varattno 13
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 13
:location 342
}
)
:location 340
}
)
:location 323
}
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location 251
}
:resno 1
:resname l_orderkey
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 1
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 2
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 2
:location 251
}
:resno 2
:resname l_partkey
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 2
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 3
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 3
:location 251
}
:resno 3
:resname l_supkey
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 3
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 4
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 4
:location 251
}
:resno 4
:resname l_linenumber
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 4
:resjunk false
}

```

```

}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 5
:vartype 701
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 5
:location 251
}
:resno 5
:resname l_quantity
:resortgroupref 0
:resorigtbl 16566
:resorigcol 5
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 6
:vartype 701
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 6
:location 251
}
:resno 6
:resname l_extendedprice
:resortgroupref 0
:resorigtbl 16566
:resorigcol 6
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 7
:vartype 701
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 7
:location 251
}
:resno 7
:resname l_discount
:resortgroupref 0
:resorigtbl 16566
:resorigcol 7
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 8
:vartype 701
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 8
:location 251
}
:resno 8
:resname l_tax
:resortgroupref 0
:resorigtbl 16566
:resorigcol 8
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 9
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 9
:location 251
}
:resno 9
:resname l_returnflag
:resortgroupref 0
:resorigtbl 16566
:resorigcol 9
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 10
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 10
:location 251
}
:resno 10
:resname l_linestatus
:resortgroupref 0
:resorigtbl 16566
:resorigcol 10
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 11
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 11
:location 251
}
}

```

```

:resno 11
:resname l_shipdate
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 11
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 12
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 12
:location 251
}
:resno 12
:resname l_commitdate
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 12
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 13
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 13
:location 251
}
:resno 13
:resname l_receiptdate
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 13
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 14
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 14
:location 251
}
:resno 14
:resname l_shipinstruct
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 14
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 15
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 15
:location 251
}
:resno 15
:resname l_shipmode
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 15
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 16
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 16
:location 251
}
:resno 16
:resname l_comment
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 16
:resjunk false
}
)
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location 0
:stmt_len 0
}
:location 227
}
}
:location 162
}
}
}
:mergeActionList <>
:mergeTargetRelation 0

```

```

:mergeJoinCondition <>
:targetList (
  {TARGETENTRY
    :expr
    {VAR
      :varno 2
      :varattno 1
      :vartype 25
      :vartypmod -1
      :varcollid 160
      :varnullingrels (b)
      :varlevelsup 0
      :varreturningtype 0
      :varnosyn 2
      :varattnosyn 1
      :location -1
    }
    :resno 1
    :resname o_orderpriority
    :ressortgroupref 1
    :resorigtbl 16847
    :resorigcol 6
    :resjunk false
  }
  {TARGETENTRY
    :expr
    {AGGREF
      :aggfnoid 2803
      :aggtype 20
      :aggcollid 0
      :inputcollid 0
      :aggtranstype 0
      :aggargtypes <>
      :aggdirectargs <>
      :args <>
      :aggorder <>
      :aggdistinct <>
      :aggfilter <>
      :aggstar true
      :aggvariadic false
      :aggkind n
      :aggpresorted false
      :agglevelsup 0
      :aggsplit 0
      :aggno -1
      :aggtransno -1
      :location 68
    }
    :resno 2
    :resname order_count
    :ressortgroupref 0
    :resorigtbl 0
    :resorigcol 0
    :resjunk false
  }
)
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause (
  {SORTGROUPCLAUSE
    :tleSortGroupRef 1
    :eqop 98
    :sortop 664
    :reverse_sort false
    :nulls_first false
    :hashable true
  }
)
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause (
  {SORTGROUPCLAUSE
    :tleSortGroupRef 1
    :eqop 98
    :sortop 664
    :reverse_sort false
    :nulls_first false
    :hashable true
  }
)
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location 0
:stmt_len 0
}
:options (
  {DEFELEM
    :defnamespace <>
    :defname analyze
    :arg <>
    :defaction 0
    :location 9
  }
  {DEFELEM
    :defnamespace <>
    :defname verbose
    :arg <>
    :defaction 0
    :location 18
  }
  {DEFELEM
    :defnamespace <>
    :defname buffers
    :arg <>
    :defaction 0
    :location 27
  }
)
}
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:cteList <>
:rtable <>
:rtepermInfos <>
:jointree <>
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList <>
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>

```

```

:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location 0
:stmt_len 419
}

```

WARNING: ===== REWRITE INPUT =====

```

DETAIL: {QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs true
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks true
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE true
:isReturn false
:cteList <>
:rtable (
{RANGETBENTRY
:alias <>
:eref
{ALIAS
:aliasname orders_secure
:colnames ("o_orderkey" "o_custkey" "o_orderstatus" "o_totalprice" "o_orderdate" "o_orderpriority" "o_clerk" "o_shippriority" "o_comment")
}
:rtekind 0
:relid 16847
:inh true
:relkind v
:rellockmode 1
:perminfoindex 1
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
{RANGETBENTRY
:alias <>
:eref
{ALIAS
:aliasname *GROUP*
:colnames ("o_orderpriority")
}
:rtekind 9
:groupexprs (
{VAR
:varno 1
:varattno 6
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 6
:location 47
}
)
:lateral false
:inFromCl false
:securityQuals <>
}
)
:rteperminfos (
{RTEPERMISSIONINFO
:relid 16847
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 12 13)
:insertedCols (b)
:updatedCols (b)
}
)
:jointree
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtable 1
}
)
:quals
{BOOLEXP
:boolop and
:args (
{OPEXP
:opno 1098
:opfuncid 1090
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 5
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 5
:location 125
}
{CONST
:consttype 1082
:consttypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location 145
:constvalue 4 [ 113 -9 -1 -1 -1 -1 -1 ]
}
)
:location 137
}
{OPEXP
:opno 2345
:opfuncid 2338
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1

```

```

:varattno 5
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 5
:location 166
}
{OPEXPR
:opno 1076
:opfuncid 2071
:opresulttype 1114
:opretset false
:opcollid 0
:inputcollid 0
:args (
{CONST
:consttype 1082
:consttypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location 185
:constvalue 4 [ 113 -9 -1 -1 -1 -1 -1 ]
}
{CONST
:consttype 1186
:consttypmod 196607
:constcollid 0
:constlen 16
:constbyval false
:constisnull false
:location 209
:constvalue 16 [ 0 0 0 0 0 0 0 0 0 0 0 0 3 0 0 0 ]
}
)
:location 198
}
)
:location 178
}
{SUBLINK
:subLinkType 0
:subLinkId 0
:testexpr <>
:operName <>
:subselect
{QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:ctelist <>
:rttable (
{RANGETBLENTRY
:alias <>
:eref
{ALIAS
:aliasname lineitem
:colnames ("l_orderkey" "l_partkey" "l_suppkey" "l_linenu
mber" "l_quantity" "l_extendedprice" "l_discount" "l_tax"
"l_returnflag" "l_linestatus" "l_shipdate" "l_commitdate
" "l_receiptdate" "l_shipinstruct" "l_shipmode" "l_commen
t")
}
:rtkind 0
:relid 16566
:inh true
:relkind r
:rellockmode 1
:perminfoindex 1
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
)
)
:rteperminfos (
{RTEPERMISSIONINFO
:relid 16566
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22
23)
:insertedCols (b)
:updatedCols (b)
}
)
)
:jointree
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 1
}
)
)
:quals
{BOOLEXPR
:boolop and
:args (
{OPEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location 289
}
)
{VAR
:varno 1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 1
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
}
)
}
}

```

```

        :location 302
    }
}
:location 300
}
{OPEXPR
:opno 1095
:opfuncid 1087
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
    {VAR
        :varno 1
        :varattno 12
        :vartype 1082
        :vartypmod -1
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 1
        :varattnosyn 12
        :location 327
    }
    {VAR
        :varno 1
        :varattno 13
        :vartype 1082
        :vartypmod -1
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 1
        :varattnosyn 13
        :location 342
    }
}
:location 340
}
}
:location 323
}
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
    {TARGETENTRY
        :expr
        {VAR
            :varno 1
            :varattno 1
            :vartype 23
            :vartypmod -1
            :varcollid 0
            :varnullingrels (b)
            :varlevelsup 0
            :varreturningtype 0
            :varnosyn 1
            :varattnosyn 1
            :location 251
        }
    }
    :resno 1
    :resname l_orderkey
    :ressortgroupref 0
    :resorigtbl 16566
    :resorigcol 1
    :resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 2
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 2
:location 251
}
:resno 2
:resname l_partkey
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 2
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 3
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 3
:location 251
}
:resno 3
:resname l_suppkey
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 3
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 4
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 4
:location 251
}
:resno 4
:resname l_linenumbers
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 4
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 5
:vartype 701
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0

```

```

:varreturningtype 0
:varnosyn 1
:varattnosyn 5
:location 251
}
:resno 5
:resname 1_quantity
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 5
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 6
:vartype 701
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 6
:location 251
}
:resno 6
:resname 1_extendedprice
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 6
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 7
:vartype 701
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 7
:location 251
}
:resno 7
:resname 1_discount
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 7
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 8
:vartype 701
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 8
:location 251
}
:resno 8
:resname 1_tax
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 8
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 9
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 9
:location 251
}
:resno 9
:resname 1_returnflag
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 9
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 10
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 10
:location 251
}
:resno 10
:resname 1_linestatus
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 10
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 11
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 11
:location 251
}
:resno 11
:resname 1_shipdate
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 11
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1

```

```

:varattno 12
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 12
:location 251
}
:resno 12
:resname 1_commitdate
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 12
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 13
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 13
:location 251
}
:resno 13
:resname 1_receiptdate
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 13
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 14
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 14
:location 251
}
:resno 14
:resname 1_shipinstruct
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 14
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 15
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 15
:location 251
}
:resno 15
:resname 1_shipmode
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 15
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 16
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 16
:location 251
}
:resno 16
:resname 1_comment
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 16
:resjunk false
}
}
:location 227
}
}
:location 162
}
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
{TARGETENTRY
:expr
{VAR
:varno 2
:varattno 1
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)

```

```

:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 1
:location -1
}
:resno 1
:resname o_orderpriority
:ressortgroupref 1
:resorigtbl 16847
:resorigcol 6
:resjunk false
}
{TARGETENTRY
:expr
{AGGREF
:aggfnoid 2803
:aggtype 28
:aggcollid 0
:inputcollid 0
:aggtranstype 0
:aggargtypes <>
:aggdirectargs <>
:args <>
:aggorder <>
:aggdistinct <>
:aggfilter <>
:aggstar true
:aggvariadic false
:aggkind n
:aggresorted false
:agglevelsup 0
:aggsplit 0
:aggno -1
:aggtransno -1
:location 68
}
:resno 2
:resname order_count
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
}
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause (
{SORTGROUPCLAUSE
:tleSortGroupRef 1
:eqop 98
:sortop 664
:reverse_sort false
:nulls_first false
:hashable true
}
)
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause (
{SORTGROUPCLAUSE
:tleSortGroupRef 1
:eqop 98
:sortop 664
:reverse_sort false
:nulls_first false
:hashable true
}
)
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location 0
:stmt_len 0
}

```

WARNING: ===== REWRITE OUTPUT =====

```

DETAIL: (
{QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs true
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks true
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE true
:isReturn false
:cteList <>
:rtable (
{RANGETBENTRY
:alias <>
:eref
{ALIAS
:aliasname orders_secure
:colnames ("o_orderkey" "o_custkey" "o_orderstatus" "o_totalprice" "o_orderdate" "o_orderpriority" "o_clerk" "o_shippriority" "o_comment")
}
}
:rtekind 1
:subquery
{QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks true
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:cteList <>
:rtable (
{RANGETBENTRY
:alias
{ALIAS
:aliasname o
:colnames <>
}
}
:eref
{ALIAS
:aliasname o
:colnames ("o_orderkey" "o_custkey" "o_orderstatus" "o_totalprice" "o_orderdate" "o_orderpriority" "o_clerk" "o_shippriority" "o_comment")
}
}
}
)

```

```

    }
    :rtekind 0
    :relid 16561
    :inh true
    :relkind r
    :rellockmode 1
    :perminfoindex 1
    :tablesample <>
    :lateral false
    :inFromCl true
    :securityQuals <>
  }
)
:rteperminfos (
  {RTEPERMISSIONINFO
    :relid 16561
    :inh true
    :requiredPerms 2
    :checkAsUser 10
    :selectedCols (b 8 9 10 11 12 13 14 15 16)
    :insertedCols (b)
    :updatedCols (b)
  }
)
:jointree
{FROMEXPR
  :fromlist (
    {RANGETBLREF
      :rtindex 1
    }
  )
  :quals
  {SUBLINK
    :subLinkType 0
    :subLinkId 0
    :testexpr <>
    :operName <>
    :subselect
    {QUERY
      :commandType 1
      :querySource 0
      :canSetTag true
      :utilityStmt <>
      :resultRelation 0
      :hasAggs false
      :hasWindowFuncs false
      :hasTargetSRFs false
      :hasSublinks false
      :hasDistinctOn false
      :hasRecursive false
      :hasModifyingCTE false
      :hasForUpdate false
      :hasRowSecurity false
      :hasGroupRTE false
      :isReturn false
      :cteList <>
      :rtable (
        {RANGETBLENTRY
          :alias
          {ALIAS
            :aliasname c
            :colnames <>
          }
          :eref
          {ALIAS
            :aliasname c
            :colnames ("c_custkey" "c_name" "c_address" "c_nationk
              ey" "c_phone" "c_acctbal" "c_mktsegment" "c_comment")
          }
          :rtekind 0
          :relid 16546
          :inh true
          :relkind r
          :rellockmode 1
          :perminfoindex 1
          :tablesample <>
          :lateral false
          :inFromCl true
          :securityQuals <>
        }
      )
    }
  }
  :rteperminfos (
    {RTEPERMISSIONINFO
      :relid 16546
      :inh true
      :requiredPerms 2
      :checkAsUser 10
      :selectedCols (b 8 13)
      :insertedCols (b)
      :updatedCols (b)
    }
  )
}
:jointree
{FROMEXPR
  :fromlist (
    {RANGETBLREF
      :rtindex 1
    }
  )
  :quals
  {BOOLEXP
    :boolop and
    :args (
      {OPEXP
        :opno 96
        :opfuncid 65
        :opresulttype 16
        :opretset false
        :opcollid 0
        :inputcollid 0
        :args (
          {VAR
            :varno 1
            :varattno 1
            :vartype 23
            :vartypmod -1
            :varcollid 0
            :varnullingrels (b)
            :varlevelsup 0
            :varreturningtype 0
            :varnosyn 1
            :varattnosyn 1
            :location -1
          }
          {VAR
            :varno 1
            :varattno 2
            :vartype 23
            :vartypmod -1
            :varcollid 0
            :varnullingrels (b)
            :varlevelsup 1
            :varreturningtype 0
            :varnosyn 1
            :varattnosyn 2
            :location -1
          }
        )
      }
    )
    :location -1
  }
  {OPEXP
    :opno 674
    :opfuncid 297
    :opresulttype 16
    :opretset false
    :opcollid 0
    :inputcollid 0
    :args (

```

```

{VAR
:varno 1
:varattno 6
:varatype 701
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 6
:location -1
}
{FUNCEXP
:funcid 316
:funcresulttype 701
:funcresultset false
:funcvariadic false
:funcformat 2
:location -1
:funccollid 0
:inputcollid 0
:args (
{CONST
:consttype 23
:consttypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location -1
:constvalue 4 [ -120 19 0 0 0 0 0 ]
}
)
:location -1
}
}
:location -1
}
}
:location -1
}
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
{TARGETENTRY
:expr
{CONST
:consttype 23
:consttypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location -1
:constvalue 4 [ 1 0 0 0 0 0 0 ]
}
:resno 1
:resname ?column?
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
)
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location -1
:stmt_len -1
}
:location -1
}
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 1
:varatype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location -1
}
:resno 1
:resname o_orderkey
:ressortgroupref 0
:resorigtbl 16561
:resorigcol 1
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 2
:varatype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 2
:location -1
}
:resno 2
:resname o_custkey
:ressortgroupref 0
:resorigtbl 16561
:resorigcol 2
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 3
:varatype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)

```

```

:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 3
:location -1
}
:resno 3
:resname o_orderstatus
:ressortgroupref 0
:resorigtbl 16561
:resorigcol 3
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 4
:varstype 701
:varstpmo -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 4
:location -1
}
:resno 4
:resname o_totalprice
:ressortgroupref 0
:resorigtbl 16561
:resorigcol 4
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 5
:varstype 1082
:varstpmo -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 5
:location -1
}
:resno 5
:resname o_orderdate
:ressortgroupref 0
:resorigtbl 16561
:resorigcol 5
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 6
:varstype 25
:varstpmo -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 6
:location -1
}
:resno 6
:resname o_orderpriority
:ressortgroupref 0
:resorigtbl 16561
:resorigcol 6
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 7
:varstype 25
:varstpmo -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 7
:location -1
}
:resno 7
:resname o_clerk
:ressortgroupref 0
:resorigtbl 16561
:resorigcol 7
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 8
:varstype 23
:varstpmo -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 8
:location -1
}
:resno 8
:resname o_shippriority
:ressortgroupref 0
:resorigtbl 16561
:resorigcol 8
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 9
:varstype 25
:varstpmo -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 9
:location -1
}
:resno 9
:resname o_comment
:ressortgroupref 0
:resorigtbl 16561
:resorigcol 9
:resjunk false
}
}
:override 0
:onConflict <>

```

```

: returningOldAlias <>
: returningNewAlias <>
: returningList <>
: groupClause <>
: groupDistinct false
: groupingSets <>
: havingQual <>
: windowClause <>
: distinctClause <>
: sortClause <>
: limitOffset <>
: limitCount <>
: limitOption 0
: rowMarks <>
: setOperations <>
: constraintDecls <>
: withCheckOptions <>
: stmt_location -1
: stmt_len -1
}
: security_barrier true
: relid 16847
: inh false
: relkind v
: reloccmode 1
: perminfoindex 1
: lateral false
: inFromCl true
: securityQuals <>
}
(RANGEABLEENTRY
: alias <>
: eref
{ALIAS
: aliasname *GROUP*
: colnames ("o_orderpriority")
}
: rtekind 9
: groupexprs (
{VAR
: varno 1
: varattno 6
: vartype 25
: vartypmod -1
: varcollid 100
: varnullingrels (b)
: varlevelsup 0
: varreturningtype 0
: varnosyn 1
: varattnosyn 6
: location 47
}
)
: lateral false
: inFromCl false
: securityQuals <>
}
)
: rtepermissions (
{RTEPERMISSIONINFO
: relid 16847
: inh true
: requiredPerms 2
: checkAsUser 0
: selectedCols (b 0 12 13)
: insertedCols (b)
: updatedCols (b)
}
)
)
: jointree
{FROMEXPR
: fromlist (
{RANGEABLEREF
: rtindex 1
}
)
: quals
{BOOLEXP
: boolop and
: args (
{OPEXP
: opno 1098
: opfuncid 1090
: opresulttype 16
: opretset false
: opcollid 0
: inputcollid 0
: args (
{VAR
: varno 1
: varattno 5
: vartype 1082
: vartypmod -1
: varcollid 0
: varnullingrels (b)
: varlevelsup 0
: varreturningtype 0
: varnosyn 1
: varattnosyn 5
: location 125
}
{CONST
: consttype 1082
: consttypmod -1
: constcollid 0
: constlen 4
: constbyval true
: constisnull false
: location 145
: constvalue 4 [ 113 -9 -1 -1 -1 -1 -1 ]
}
)
: location 137
}
{OPEXP
: opno 2345
: opfuncid 2338
: opresulttype 16
: opretset false
: opcollid 0
: inputcollid 0
: args (
{VAR
: varno 1
: varattno 5
: vartype 1082
: vartypmod -1
: varcollid 0
: varnullingrels (b)
: varlevelsup 0
: varreturningtype 0
: varnosyn 1
: varattnosyn 5
: location 166
}
{OPEXP
: opno 1076
: opfuncid 2071
: opresulttype 1114
: opretset false
: opcollid 0
: inputcollid 0
: args (
{CONST
: consttype 1082
: consttypmod -1
: constcollid 0
: constlen 4

```

```

:constbyval true
:constisnull false
:location 185
:constvalue 4 [ 113 -9 -1 -1 -1 -1 -1 ]
}
{CONST
:consttype 1186
:consttypmod 196607
:constcollid 0
:constlen 16
:constbyval false
:constisnull false
:location 209
:constvalue 16 [ 0 0 0 0 0 0 0 0 0 0 0 0 3 0 0 0 ]
}
}
:location 198
}
)
:location 178
}
{SUBLINK
:sublinkType 0
:sublinkId 0
:testexpr <>
:operName <>
:subselect
{QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:cteList <>
:rtable (
{RANGETBLENTRY
:alias <>
:eref
{ALIAS
:aliasname lineitem
:colnames ("l_orderkey" "l_partkey" "l_suppkey" "l_linenu
mber" "l_quantity" "l_extendedprice" "l_discount" "l_tax"
"l_returnflag" "l_linestatus" "l_shipdate" "l_commitdate
" "l_receiptdate" "l_shipinstruct" "l_shipmode" "l_commen
t")
}
:rtekind 0
:relid 16566
:inh true
:relkind r
:relockmode 1
:perminfoindex 1
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
)
:rteperinfos (
{RTEPERMISSIONINFO
:relid 16566
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22
23)
:insertedCols (b)
:updatedCols (b)
}
)
):jointree
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 1
}
)
):quals
{BOOLEXP
:boolop and
:args (
{OPEXP
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location 289
}
{VAR
:varno 1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 1
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location 302
}
}
}
:location 300
}
{OPEXP
:opno 1095
:opfuncid 1087
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 12
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 12

```

```

        :location 327
      }
      {VAR
        :varno 1
        :varattno 13
        :vartype 1082
        :vartypmod -1
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 1
        :varattnosyn 13
        :location 342
      }
    }
    :location 340
  }
)
:location 323
}
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
  {TARGETENTRY
    :expr
    {VAR
      :varno 1
      :varattno 1
      :vartype 23
      :vartypmod -1
      :varcollid 0
      :varnullingrels (b)
      :varlevelsup 0
      :varreturningtype 0
      :varnosyn 1
      :varattnosyn 1
      :location 251
    }
    :resno 1
    :resname 1_orderkey
    :ressortgroupref 0
    :resorigtbl 16566
    :resorigcol 1
    :resjunk false
  }
  {TARGETENTRY
    :expr
    {VAR
      :varno 1
      :varattno 2
      :vartype 23
      :vartypmod -1
      :varcollid 0
      :varnullingrels (b)
      :varlevelsup 0
      :varreturningtype 0
      :varnosyn 1
      :varattnosyn 2
      :location 251
    }
    :resno 2
    :resname 1_partkey
    :ressortgroupref 0
    :resorigtbl 16566
    :resorigcol 2
    :resjunk false
  }
  {TARGETENTRY
    :expr
    {VAR
      :varno 1
      :varattno 3
      :vartype 23
      :vartypmod -1
      :varcollid 0
      :varnullingrels (b)
      :varlevelsup 0
      :varreturningtype 0
      :varnosyn 1
      :varattnosyn 3
      :location 251
    }
    :resno 3
    :resname 1_supplekey
    :ressortgroupref 0
    :resorigtbl 16566
    :resorigcol 3
    :resjunk false
  }
  {TARGETENTRY
    :expr
    {VAR
      :varno 1
      :varattno 4
      :vartype 23
      :vartypmod -1
      :varcollid 0
      :varnullingrels (b)
      :varlevelsup 0
      :varreturningtype 0
      :varnosyn 1
      :varattnosyn 4
      :location 251
    }
    :resno 4
    :resname 1_linenummer
    :ressortgroupref 0
    :resorigtbl 16566
    :resorigcol 4
    :resjunk false
  }
  {TARGETENTRY
    :expr
    {VAR
      :varno 1
      :varattno 5
      :vartype 701
      :vartypmod -1
      :varcollid 0
      :varnullingrels (b)
      :varlevelsup 0
      :varreturningtype 0
      :varnosyn 1
      :varattnosyn 5
      :location 251
    }
    :resno 5
    :resname 1_quantity
    :ressortgroupref 0
    :resorigtbl 16566
    :resorigcol 5
    :resjunk false
  }
  {TARGETENTRY
    :expr
    {VAR
      :varno 1
      :varattno 6
      :vartype 701
      :vartypmod -1
      :varcollid 0
      :varnullingrels (b)
      :varlevelsup 0
      :varreturningtype 0
      :varnosyn 1

```

```

:varattnosyn 6
:location 251
}
:resno 6
:resname 1_extendedprice
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 6
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 7
:vartype 701
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 7
:location 251
}
:resno 7
:resname 1_discount
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 7
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 8
:vartype 701
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 8
:location 251
}
:resno 8
:resname 1_tax
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 8
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 9
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 9
:location 251
}
:resno 9
:resname 1_returnflag
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 9
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 10
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 10
:location 251
}
:resno 10
:resname 1_linestatus
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 10
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 11
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 11
:location 251
}
:resno 11
:resname 1_shipdate
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 11
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 12
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 12
:location 251
}
:resno 12
:resname 1_commitdate
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 12
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 13
:vartype 1082

```

```

:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 13
:location 251
}
:resno 13
:resname l_receiptdate
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 13
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 14
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 14
:location 251
}
:resno 14
:resname l_shipinstruct
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 14
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 15
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 15
:location 251
}
:resno 15
:resname l_shipmode
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 15
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 16
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 16
:location 251
}
:resno 16
:resname l_comment
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 16
:resjunk false
}
}
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location 0
:stmt_len 0
}
:location 227
}
}
:location 162
}
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
{TARGETENTRY
:expr
{VAR
:varno 2
:varattno 1
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 1
:location -1
}
:resno 1
:resname o_orderpriority
:ressortgroupref 1
:resorigtbl 16847
:resorigcol 6
:resjunk false
}
{TARGETENTRY
:expr
{AGGREF
:aggfnoid 2803
:aggtype 20
:aggcollid 0
:inputcollid 0
:aggtranstype 0
:aggargtypes <>
:aggdirectargs <>
:args <>
}
}

```

```

:aggorder <>
:aggdistinct <>
:aggfilter <>
:aggstar true
:aggvariadic false
:aggkind n
:aggpresorted false
:agglevelsup 0
:aggsplit 0
:aggno -1
:aggtransno -1
:location 68
}
:resno 2
:resname order_count
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
)
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause (
  {SORTGROUPCLAUSE
    :tleSortGroupRef 1
    :eqop 98
    :sortop 664
    :reverse_sort false
    :nulls_first false
    :hashable true
  }
)
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause (
  {SORTGROUPCLAUSE
    :tleSortGroupRef 1
    :eqop 98
    :sortop 664
    :reverse_sort false
    :nulls_first false
    :hashable true
  }
)
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location 0
:stmt_len 0
}
)

```

WARNING: ===== PLANNER INPUT =====

```

DETAIL: {QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs true
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks true
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE true
:isReturn false
:cteList <>
:rtable (
  {RANGETBENTRY
    :alias <>
    :eref
      {ALIAS
        :aliasname orders_secure
        :colnames ("o_orderkey" "o_custkey" "o_orderstatus" "o_totalprice" "o_orderdate" "o_orderpriority" "o_clerk" "o_shippriority" "o_comment")
      }
    :rtekind 1
    :subquery
      {QUERY
        :commandType 1
        :querySource 0
        :canSetTag true
        :utilityStmt <>
        :resultRelation 0
        :hasAggs false
        :hasWindowFuncs false
        :hasTargetSRFs false
        :hasSubLinks true
        :hasDistinctOn false
        :hasRecursive false
        :hasModifyingCTE false
        :hasForUpdate false
        :hasRowSecurity false
        :hasGroupRTE false
        :isReturn false
        :cteList <>
       :rtable (
          {RANGETBENTRY
            :alias
              {ALIAS
                :aliasname o
                :colnames <>
              }
            :eref
              {ALIAS
                :aliasname o
                :colnames ("o_orderkey" "o_custkey" "o_orderstatus" "o_totalprice" "o_orderdate" "o_orderpriority" "o_clerk" "o_shippriority" "o_comment")
              }
            :rtekind 0
            :relid 16561
            :inh true
            :relkind r
            :rellockmode 1
            :perminfoindex 1
            :tablesample <>
            :lateral false
            :inFromCl true
            :securityQuals <>
          }
        )
        :rteperminfos (
          {RTEPERMISSIONINFO
            :relid 16561
            :inh true
            :requiredPerms 2
            :checkAsUser 10
            :selectedCols (b 8 9 10 11 12 13 14 15 16)
            :insertedCols (b)
            :updatedCols (b)
          }
        )
      }
    }
  )
}

```

```

:jointree
{FROMEXPR
:fromlist {
{RANGETBLREF
:rtindex 1
}
}
:quals
{SUBLINK
:subLinkType 0
:subLinkId 0
:testexpr <>
:operName <>
:subselect
{QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSublinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:cteList <>
:rtable {
{RANGETBLENTRY
:alias
{ALIAS
:aliasname c
:colnames <>
}
:eref
{ALIAS
:aliasname c
:colnames ("c_custkey" "c_name" "c_address" "c_nationk
ey" "c_phone" "c_acctbal" "c_mktsegment" "c_comment")
}
:rtekind 0
:relid 16546
:inh true
:relkind r
:rellockmode 1
:perminfoindex 1
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
}
:rteperminfos {
{RTEPERMISSIONINFO
:relid 16546
:inh true
:requiredPerms 2
:checkAsUser 10
:selectedCols (b 8 13)
:insertedCols (b)
:updatedCols (b)
}
}
}
:jointree
{FROMEXPR
:fromlist {
{RANGETBLREF
:rtindex 1
}
}
:quals
{BOOLEXP
:boolop and
:args {
{OPEXP
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args {
{VAR
:varno 1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location -1
}
}
{VAR
:varno 1
:varattno 2
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 1
:varreturningtype 0
:varnosyn 1
:varattnosyn 2
:location -1
}
}
}
:location -1
}
{OPEXP
:opno 674
:opfuncid 297
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args {
{VAR
:varno 1
:varattno 6
:vartype 701
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 6
:location -1
}
}
{FUNCEXP
:funcid 316
:funcresulttype 701
:funcset false
:funcvariadic false
:funcformat 2
:funccollid 0
:inputcollid 0
:args {
{CONST
:consttype 23

```

```

        :consttypmod -1
        :constcollid 0
        :constlen 4
        :constbyval true
        :constisnull false
        :location -1
        :constvalue 4 [ -120 19 0 0 0 0 0 0 ]
    }
    }
    :location -1
    }
    }
    :location -1
    }
    }
    :location -1
    }
    }
    :mergeActionList <>
    :mergeTargetRelation 0
    :mergeJoinCondition <>
    :targetList (
    {TARGETENTRY
    :expr
    {CONST
    :consttype 23
    :consttypmod -1
    :constcollid 0
    :constlen 4
    :constbyval true
    :constisnull false
    :location -1
    :constvalue 4 [ 1 0 0 0 0 0 0 0 ]
    }
    :resno 1
    :resname ?column?
    :ressortgrouppref 0
    :resorigtbl 0
    :resorigcol 0
    :resjunk false
    }
    )
    :override 0
    :onConflict <>
    :returningOldAlias <>
    :returningNewAlias <>
    :returningList <>
    :groupClause <>
    :groupDistinct false
    :groupingSets <>
    :havingQual <>
    :windowClause <>
    :distinctClause <>
    :sortClause <>
    :limitOffset <>
    :limitCount <>
    :limitOption 0
    :rowMarks <>
    :setOperations <>
    :constraintDeps <>
    :withCheckOptions <>
    :stmt_location -1
    :stmt_len -1
    }
    :location -1
    }
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location -1
}
:resno 1
:resname o_orderkey
:ressortgrouppref 0
:resorigtbl 16561
:resorigcol 1
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 2
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 2
:location -1
}
:resno 2
:resname o_custkey
:ressortgrouppref 0
:resorigtbl 16561
:resorigcol 2
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 3
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 3
:location -1
}
:resno 3
:resname o_orderstatus
:ressortgrouppref 0
:resorigtbl 16561
:resorigcol 3
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 4
:vartype 701
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0

```

```

:varnosyn 1
:varattnosyn 4
:location -1
}
:resno 4
:resname o_totalprice
:resortgroupref 0
:resorigtbl 16561
:resorigcol 4
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 5
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 5
:location -1
}
:resno 5
:resname o_orderdate
:resortgroupref 0
:resorigtbl 16561
:resorigcol 5
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 6
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 6
:location -1
}
:resno 6
:resname o_orderpriority
:resortgroupref 0
:resorigtbl 16561
:resorigcol 6
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 7
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 7
:location -1
}
:resno 7
:resname o_clerk
:resortgroupref 0
:resorigtbl 16561
:resorigcol 7
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 8
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 8
:location -1
}
:resno 8
:resname o_shippriority
:resortgroupref 0
:resorigtbl 16561
:resorigcol 8
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 9
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 9
:location -1
}
:resno 9
:resname o_comment
:resortgroupref 0
:resorigtbl 16561
:resorigcol 9
:resjunk false
}
}
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location -1
:stmt_len -1
}
:security_barrier true
:relid 16847
:inh false
:relkind v

```

```

:rellockmode 1
:perminfoindex 1
:lateral false
:inFromCl true
:securityQuals <>
}
{RANGETBENTRY
:alias <>
:eref
{ALIAS
:aliasname *GROUP*
:colnames ("o_orderpriority")
}
:rtekind 9
:groupexprs (
{VAR
:varno 1
:varattno 6
:varatype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 6
:location 47
}
)
:lateral false
:inFromCl false
:securityQuals <>
}
)
:rteperminfos (
{RTEPERMISSIONINFO
:relid 16847
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 12 13)
:insertedCols (b)
:updatedCols (b)
}
)
:jointree
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtdindex 1
}
)
}
:quals
{BOOLEXP
:boolop and
:args (
{OEXPR
:opno 1098
:opfuncid 1090
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 5
:varatype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 5
:location 125
}
{CONST
:consttype 1082
:consttypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location 145
:constvalue 4 [ 113 -9 -1 -1 -1 -1 -1 -1 ]
}
)
}
:location 137
}
{OEXPR
:opno 2345
:opfuncid 2338
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 5
:varatype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 5
:location 166
}
{OEXPR
:opno 1076
:opfuncid 2071
:opresulttype 1114
:opretset false
:opcollid 0
:inputcollid 0
:args (
{CONST
:consttype 1082
:consttypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location 185
:constvalue 4 [ 113 -9 -1 -1 -1 -1 -1 -1 ]
}
{CONST
:consttype 1186
:consttypmod 196607
:constcollid 0
:constlen 16
:constbyval false
:constisnull false
:location 209
:constvalue 16 [ 0 0 0 0 0 0 0 0 0 0 0 0 0 3 0 0 0 ]
}
)
}
:location 198
}
)
:location 178
}
{SUBLINK
:subLinkType 0
:subLinkId 0
}

```

```

:testexpr <>
:operName <>
:subselect
{QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:cteList <>
:rtable (
{RANGETBLENTRY
:alias <>
:eref
{ALIAS
:aliasname lineitem
:colnames ("l_orderkey" "l_partkey" "l_suppkey" "l_linenu
mber" "l_quantity" "l_extendedprice" "l_discount" "l_tax"
"l_returnflag" "l_linestatus" "l_shipdate" "l_commitdate
" "l_receiptdate" "l_shipinstruct" "l_shipmode" "l_commen
t")
}
:rtekind 0
:relid 16566
:inh true
:relkind r
:rellockmode 1
:perminfoindex 1
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
)
:rteperinfos (
{RTEPERMISSIONINFO
:relid 16566
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22
23)
:insertedCols (b)
:updatedCols (b)
}
)
:jointree
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtable 1
}
)
:quals
{BOOLEXP
:boolop and
:args (
{OPEXP
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location 289
}
{VAR
:varno 1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 1
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location 302
}
}
:location 300
}
{OPEXP
:opno 1095
:opfuncid 1087
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 12
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 12
:location 327
}
{VAR
:varno 1
:varattno 13
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 13
:location 342
}
}
:location 340
}
}
:location 323
}
}
:mergeActionList <>
:mergeTargetRelation 0

```

```

:mergeJoinCondition <>
:targetList (
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location 251
}
:resno 1
:resname l_orderkey
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 1
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 2
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 2
:location 251
}
:resno 2
:resname l_partkey
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 2
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 3
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 3
:location 251
}
:resno 3
:resname l_supplekey
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 3
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 4
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 4
:location 251
}
:resno 4
:resname l_linenummer
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 4
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 5
:vartype 701
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 5
:location 251
}
:resno 5
:resname l_quantity
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 5
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 6
:vartype 701
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 6
:location 251
}
:resno 6
:resname l_extendedprice
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 6
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 7
:vartype 701
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 7
:location 251
}

```

```

    }
:resno 7
:resname l_discount
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 7
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 8
:vartype 701
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 8
:location 251
}
:resno 8
:resname l_tax
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 8
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 9
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 9
:location 251
}
:resno 9
:resname l_returnflag
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 9
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 10
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 10
:location 251
}
:resno 10
:resname l_linestatus
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 10
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 11
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 11
:location 251
}
:resno 11
:resname l_shipdate
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 11
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 12
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 12
:location 251
}
:resno 12
:resname l_commitdate
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 12
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 13
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 13
:location 251
}
:resno 13
:resname l_receiptdate
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 13
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 14
:vartype 25
:vartypmod -1
:varcollid 100

```

```

        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 1
        :varattnosyn 14
        :location 251
    }
    :resno 14
    :resname l_shipinstruct
    :ressortgroupref 0
    :resorigtbl 16566
    :resorigcol 14
    :resjunk false
    }
    {TARGETENTRY
    :expr
    {VAR
    :varno 1
    :varattno 15
    :vartype 25
    :vartypmod -1
    :varcollid 100
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 15
    :location 251
    }
    :resno 15
    :resname l_shipmode
    :ressortgroupref 0
    :resorigtbl 16566
    :resorigcol 15
    :resjunk false
    }
    {TARGETENTRY
    :expr
    {VAR
    :varno 1
    :varattno 16
    :vartype 25
    :vartypmod -1
    :varcollid 100
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 16
    :location 251
    }
    :resno 16
    :resname l_comment
    :ressortgroupref 0
    :resorigtbl 16566
    :resorigcol 16
    :resjunk false
    }
    }
    :location 227
    }
    :location 162
    }
    }
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
{TARGETENTRY
:expr
{VAR
:varno 2
:varattno 1
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 1
:location -1
}
:resno 1
:resname o_orderpriority
:ressortgroupref 1
:resorigtbl 16847
:resorigcol 6
:resjunk false
}
{TARGETENTRY
:expr
{AGGREF
:aggfnoid 2803
:aggtype 20
:aggcollid 0
:inputcollid 0
:aggtranstype 0
:aggargtypes <>
:aggdirectargs <>
:args <>
:aggorder <>
:aggdistinct <>
:aggfilter <>
:aggstar true
:aggvariadic false
:aggkind n
:aggpresorted false
:agglevelsup 0
:aggssplit 0
:aggno -1
:aggtransno -1
:location 68
}
:resno 2
:resname order_count
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
)
:override 0
:onConflict <>
:returningOldAlias <>

```

```

:returningNewAlias <>
:returningList <>
:groupClause (
  {SORTGROUPCLAUSE
    :tleSortGroupRef 1
    :eqop 98
    :sortop 664
    :reverse_sort false
    :nulls_first false
    :hashable true
  }
)
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause (
  {SORTGROUPCLAUSE
    :tleSortGroupRef 1
    :eqop 98
    :sortop 664
    :reverse_sort false
    :nulls_first false
    :hashable true
  }
)
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location 0
:stmt_len 0
)

WARNING: ===== FINAL PLAN =====
DETAIL:   {AGG
:plan.disabled_nodes 0
:plan.startup_cost 199632.93129729395
:plan.total_cost 199693.2352072957
:plan.plan_rows 200
:plan.plan_width 17
:plan.parallel_aware false
:plan.parallel_safe false
:plan.async_capable false
:plan.plan_node_id 0
:plan.targetlist (
  {TARGETENTRY
    :expr
    {VAR
      :varno -2
      :varattno 1
      :vartype 25
      :vartypmod -1
      :varcollid 100
      :varnullingrels (b)
      :varlevelsup 0
      :varreturningtype 0
      :varnosyn 0
      :varattnosyn 0
      :location -1
    }
    :resno 1
    :resname o_orderpriority
    :ressortgroupref 1
    :resorigtbl 16847
    :resorigcol 6
    :resjunk false
  }
  {TARGETENTRY
    :expr
    {AGGREF
      :aggfnoid 2003
      :aggtype 20
      :aggcollid 0
      :inputcollid 0
      :aggtranstype 20
      :aggargtypes <>
      :aggdirectargs <>
      :args (
        {TARGETENTRY
          :expr
          {VAR
            :varno -2
            :varattno 2
            :vartype 20
            :vartypmod -1
            :varcollid 0
            :varnullingrels (b)
            :varlevelsup 0
            :varreturningtype 0
            :varnosyn 0
            :varattnosyn 0
            :location -1
          }
          :resno 1
          :resname <>
          :ressortgroupref 0
          :resorigtbl 0
          :resorigcol 0
          :resjunk false
        }
      )
      :aggorder <>
      :aggdistinct <>
      :aggfilter <>
      :aggstar true
      :aggvariadic false
      :aggkind n
      :aggpresorted false
      :agglevelsup 0
      :aggssplit 9
      :aggno 0
      :aggtransno 0
      :location 68
    }
    :resno 2
    :resname order_count
    :ressortgroupref 0
    :resorigtbl 0
    :resorigcol 0
    :resjunk false
  }
)
:plan.qual <>
:plan.lefttree
  {GATHERMERGE
    :plan.disabled_nodes 0
    :plan.startup_cost 199632.93129729395
    :plan.total_cost 199688.83520729566
    :plan.plan_rows 480
    :plan.plan_width 17
    :plan.parallel_aware false
    :plan.parallel_safe false
    :plan.async_capable false
    :plan.plan_node_id 1
    :plan.targetlist (
      {TARGETENTRY
        :expr
        {VAR
          :varno -2
          :varattno 1
          :vartype 25
          :vartypmod -1
          :varcollid 100

```

```

:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 0
:varattnosyn 0
:location -1
}
:resno 1
:resname <>
:ressortgroupref 1
:resorigtbl 0
:resorigcol 0
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno -2
:varattno 2
:vartype 20
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 0
:varattnosyn 0
:location -1
}
:resno 2
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
}
:plan.qual <>
:plan.lefttree
{SORT
:plan.disabled_nodes 0
:plan.startup_cost 198632.90752285643
:plan.total_cost 198633.40752285643
:plan.plan_rows 200
:plan.plan_width 17
:plan.parallel_aware false
:plan.parallel_safe true
:plan.async_capable false
:plan.plan_node_id 2
:plan.targetlist (
{TARGETENTRY
:expr
{VAR
:varno -2
:varattno 1
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 6
:location -1
}
:resno 1
:resname <>
:ressortgroupref 1
:resorigtbl 0
:resorigcol 0
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno -2
:varattno 2
:vartype 20
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 0
:varattnosyn 0
:location -1
}
:resno 2
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
}
:plan.qual <>
:plan.lefttree
{AGG
:plan.disabled_nodes 0
:plan.startup_cost 198623.26366666667
:plan.total_cost 198625.26366666667
:plan.plan_rows 200
:plan.plan_width 17
:plan.parallel_aware false
:plan.parallel_safe true
:plan.async_capable false
:plan.plan_node_id 3
:plan.targetlist (
{TARGETENTRY
:expr
{VAR
:varno -2
:varattno 1
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 0
:varattnosyn 0
:location -1
}
:resno 1
:resname <>
:ressortgroupref 1
:resorigtbl 0
:resorigcol 0
:resjunk false
}
{TARGETENTRY
:expr
{AGGREF
:aggfnoid 2803
:aggtype 20
:aggcollid 0
:inputcollid 0
:aggtranstype 20
:aggargtypes <>
:aggdirectargs <>
:args <>
:aggorder <>
:aggdistinct <>
:aggfilter <>
:aggstar true
:aggvariadic false
:aggkind n
:aggpresorted false
}
}
}
}

```

```

:agglevelsup 0
:aggsplit 6
:aggno 0
:aggtransno 0
:location 68
}
:resno 2
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
)
:plan.qual <>
:plan.lefttree
{HASHJOIN
:join.plan.disabled_nodes 0
:join.plan.startup_cost 154455.46666666667
:join.plan.total_cost 198460.27366666668
:join.plan.plan_rows 32598
:join.plan.plan_width 9
:join.plan.parallel_aware true
:join.plan.parallel_safe true
:join.plan.async_capable false
:join.plan.plan_node_id 4
:join.plan.targetlist (
{TARGETENTRY
:expr
{VAR
:varno -2
:varattno 1
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 6
:location -1
}
:resno 1
:resname <>
:ressortgroupref 1
:resorigtbl 0
:resorigcol 0
:resjunk false
}
}
:join.plan.qual <>
:join.plan.lefttree
{SUBQUERYSCAN
:scan.plan.disabled_nodes 0
:scan.plan.startup_cost 4806.4125
:scan.plan.total_cost 44383.42724999999
:scan.plan.plan_rows 65196
:scan.plan.plan_width 13
:scan.plan.parallel_aware false
:scan.plan.parallel_safe true
:scan.plan.async_capable false
:scan.plan.plan_node_id 5
:scan.plan.targetlist (
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 6
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 6
:location -1
}
:resno 1
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location 302
}
:resno 2
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
}
)
:scan.plan.qual (
{OPEXPR
:opno 2345
:opfuncid 2338
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 5
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 5
:location 166
}
{CONST
:consttype 1114
:consttypmod -1
:constcollid 0
:constlen 8
:constbyval true
:constisnull false
:location -1
:constvalue 8 [ 0 32 105 23 -25 90 -1 -1 ]
}
)
:location 178
}
)
:scan.plan.lefttree <>
:scan.plan.righttree <>
:scan.plan.initPlan <>
:scan.plan.extParam (b 0)

```

```

:scan.plan.allParam (b 0)
:scan.scanrelid 1
:subplan
{HASHJOIN
:join.plan.disabled_nodes 0
:join.plan.startup_cost 4806.4125
:join.plan.total_cost 41938.577249999995
:join.plan.plan_rows 195588
:join.plan.plan_width 129
:join.plan.parallel_aware true
:join.plan.parallel_safe true
:join.plan.async_capable false
:join.plan.plan_node_id 6
:join.plan.targetlist (
{TARGETENTRY
:expr
{VAR
:varno -2
:varattno 1
:vartype 23
:varitypmode -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 4
:varattnosyn 1
:location -1
}
:resno 1
:resname o_orderkey
:ressortgroupref 0
:resorigtbl 16561
:resorigcol 1
:resjunk false
}
{TARGETENTRY
:expr
{CONST
:consttype 23
:constitypmode -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull true
:location -1
:constvalue <>
}
:resno 2
:resname o_custkey
:ressortgroupref 0
:resorigtbl 16561
:resorigcol 2
:resjunk false
}
{TARGETENTRY
:expr
{CONST
:consttype 25
:constitypmode -1
:constcollid 100
:constlen -1
:constbyval false
:constisnull true
:location -1
:constvalue <>
}
:resno 3
:resname o_orderstatus
:ressortgroupref 0
:resorigtbl 16561
:resorigcol 3
:resjunk false
}
{TARGETENTRY
:expr
{CONST
:consttype 701
:constitypmode -1
:constcollid 0
:constlen 8
:constbyval true
:constisnull true
:location -1
:constvalue <>
}
:resno 4
:resname o_totalprice
:ressortgroupref 0
:resorigtbl 16561
:resorigcol 4
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno -2
:varattno 5
:vartype 1082
:varitypmode -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 4
:varattnosyn 5
:location -1
}
:resno 5
:resname o_orderdate
:ressortgroupref 0
:resorigtbl 16561
:resorigcol 5
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno -2
:varattno 6
:vartype 25
:varitypmode -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 4
:varattnosyn 6
:location -1
}
:resno 6
:resname o_orderpriority
:ressortgroupref 0
:resorigtbl 16561
:resorigcol 6
:resjunk false
}
{TARGETENTRY
:expr
{CONST
:consttype 25
:constitypmode -1
:constcollid 100
:constlen -1
:constbyval false
:constisnull true
:location -1
:constvalue <>
}
}
}

```

```

    }
    :resno 7
    :resname o_clerk
    :ressortgroupref 0
    :resorigtbl 16561
    :resorigcol 7
    :resjunk false
  }
  {TARGETENTRY
  :expr
  {CONST
  :consttype 23
  :consttypmod -1
  :constcollid 0
  :constlen 4
  :constbyval true
  :constisnull true
  :location -1
  :constvalue <>
  }
  :resno 8
  :resname o_shippriority
  :ressortgroupref 0
  :resorigtbl 16561
  :resorigcol 8
  :resjunk false
  }
  {TARGETENTRY
  :expr
  {CONST
  :consttype 25
  :consttypmod -1
  :constcollid 100
  :constlen -1
  :constbyval false
  :constisnull true
  :location -1
  :constvalue <>
  }
  :resno 9
  :resname o_comment
  :ressortgroupref 0
  :resorigtbl 16561
  :resorigcol 9
  :resjunk false
  }
)
:join.plan.qual <>
:join.plan.lefttree
{SESCAN
:scan.plan.disabled_nodes 0
:scan.plan.startup_cost 0
:scan.plan.total_cost 33820.5
:scan.plan.plan_rows 432666
:scan.plan.plan_width 21
:scan.plan.parallel_aware true
:scan.plan.parallel_safe true
:scan.plan.async_capable false
:scan.plan.plan_node_id 7
:scan.plan.targetlist (
{TARGETENTRY
:expr
{VAR
:varno 4
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 4
:varattnosyn 1
:location -1
}
:resno 1
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 4
:varattno 2
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 4
:varattnosyn 2
:location -1
}
:resno 2
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 4
:varattno 3
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 4
:varattnosyn 3
:location -1
}
:resno 3
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 4
:varattno 4
:vartype 701
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 4
:varattnosyn 4
:location -1
}
:resno 4
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}

```

```

{TARGETENTRY
:expr
{VAR
:varno 4
:varattno 5
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 4
:varattnosyn 5
:location -1
}
:resno 5
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 4
:varattno 6
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 4
:varattnosyn 6
:location -1
}
:resno 6
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 4
:varattno 7
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 4
:varattnosyn 7
:location -1
}
:resno 7
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 4
:varattno 8
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 4
:varattnosyn 8
:location -1
}
:resno 8
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 4
:varattno 9
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 4
:varattnosyn 9
:location -1
}
:resno 9
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
}
:scan.plan.qual (
{OPEXPR
:opno 1098
:opfuncid 1090
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 4
:varattno 5
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 4
:varattnosyn 5
:location -1
}
{CONST
:consttype 1082
:consttypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location 145
:constvalue 4 [ 113 -9 -1 -1 -1 -1 -1 ]
}
)
:location 137
}
)
:scan.plan.lefttree <>

```

```

:scan.plan.righttree <>
:scan.plan.initPlan <>
:scan.plan.extParam (b 0)
:scan.plan.allParam (b 0)
:scan.scanreliid 4
}
:join.plan.righttree
{HASH
:plan.disabled_nodes 0
:plan.startup_cost 4453.25
:plan.total_cost 4453.25
:plan.plan_rows 28253
:plan.plan_width 4
:plan.parallel_aware true
:plan.parallel_safe true
:plan.async_capable false
:plan.plan_node_id 8
:plan.targetlist (
{TARGETENTRY
:expr
{VAR
:varno -2
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 5
:varattnosyn 1
:location -1
}
:resno 1
:resname <>
:ressortgrouppref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
)
:plan.qual <>
:plan.lefttree
{SESCAN
:scan.plan.disabled_nodes 0
:scan.plan.startup_cost 0
:scan.plan.total_cost 4453.25
:scan.plan.plan_rows 28253
:scan.plan.plan_width 4
:scan.plan.parallel_aware true
:scan.plan.parallel_safe true
:scan.plan.async_capable false
:scan.plan.plan_node_id 9
:scan.plan.targetlist (
{TARGETENTRY
:expr
{VAR
:varno 5
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 5
:varattnosyn 1
:location -1
}
:resno 1
:resname <>
:ressortgrouppref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
)
:scan.plan.qual (
{OEXPR
:opno 674
:opfuncid 297
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 5
:varattno 6
:vartype 701
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 5
:varattnosyn 6
:location -1
}
{CONST
:consttype 701
:consttypmod -1
:constcollid 0
:constlen 8
:constbyval true
:constisnull false
:location -1
:constvalue 8 [ 0 0 0 0 0 -120 -77 64 ]
}
)
:location -1
}
)
:scan.plan.lefttree <>
:scan.plan.righttree <>
:scan.plan.initPlan <>
:scan.plan.extParam (b 0)
:scan.plan.allParam (b 0)
:scan.scanreliid 5
}
:plan.righttree <>
:plan.initPlan <>
:plan.extParam (b 0)
:plan.allParam (b 0)
:hashkeys (
{VAR
:varno -2
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 5
:varattnosyn 1
:location -1
}
)
:skewTable 16561
:skewColumn 2
:skewinherit false
:rows_total 67807.2
}
:join.plan.initPlan <>
:join.plan.extParam (b 0)
:join.plan.allParam (b 0)
:join.jointype 4

```

```

:join.inner_unique false
:join.joinqual <>
:hashclauses (
  {OPEXPR
    :opno 96
    :opfuncid 65
    :opresulttype 16
    :opretset false
    :opcollid 0
    :inputcollid 0
    :args (
      {VAR
        :varno -2
        :varattno 2
        :vartype 23
        :vartypmod -1
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 4
        :varattnosyn 2
        :location -1
      }
      {VAR
        :varno -1
        :varattno 1
        :vartype 23
        :vartypmod -1
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 5
        :varattnosyn 1
        :location -1
      }
    )
    :location -1
  }
)
:hashoperators (o 96)
:hashcollations (o 0)
:hashkeys (
  {VAR
    :varno -2
    :varattno 2
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 4
    :varattnosyn 2
    :location -1
  }
)
:scanstatus 2
)
:join.plan.righttree
{HASH
  :plan.disabled_nodes 0
  :plan.startup_cost 135974.79166666666
  :plan.total_cost 135974.79166666666
  :plan.plan_rows 833461
  :plan.plan_width 4
  :plan.parallel_aware true
  :plan.parallel_safe true
  :plan.async_capable false
  :plan.plan_node_id 10
  :plan.targetlist (
    {TARGETENTRY
      :expr
      {VAR
        :varno -2
        :varattno 1
        :vartype 23
        :vartypmod -1
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 3
        :varattnosyn 1
        :location -1
      }
    }
    :resno 1
    :resname <>
    :ressortgroupref 0
    :resorigtbl 0
    :resorigcol 0
    :resjunk false
  )
)
:plan.qual <>
:plan.lefttree
{SEQSCAN
  :scan.plan.disabled_nodes 0
  :scan.plan.startup_cost 0
  :scan.plan.total_cost 135974.79166666666
  :scan.plan.plan_rows 833461
  :scan.plan.plan_width 4
  :scan.plan.parallel_aware true
  :scan.plan.parallel_safe true
  :scan.plan.async_capable false
  :scan.plan.plan_node_id 11
  :scan.plan.targetlist (
    {TARGETENTRY
      :expr
      {VAR
        :varno 3
        :varattno 1
        :vartype 23
        :vartypmod -1
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 3
        :varattnosyn 1
        :location 289
      }
    }
    :resno 1
    :resname <>
    :ressortgroupref 0
    :resorigtbl 0
    :resorigcol 0
    :resjunk false
  )
)
:scan.plan.qual (
  {OPEXPR
    :opno 1095
    :opfuncid 1087
    :opresulttype 16
    :opretset false
    :opcollid 0
    :inputcollid 0
    :args (
      {VAR
        :varno 3
        :varattno 12
        :vartype 1082
        :vartypmod -1
        :varcollid 0
        :varnullingrels (b)

```

```

        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 3
        :varattnosyn 12
        :location 327
    }
    {VAR
    :varno 3
    :varattno 13
    :varatype 1082
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 3
    :varattnosyn 13
    :location 342
    }
    }
    :location 340
}
}
:scan.plan.lefttree <>
:scan.plan.righttree <>
:scan.plan.initPlan <>
:scan.plan.extParam (b 0)
:scan.plan.allParam (b 0)
:scan.scanrelid 3
}
:plan.righttree <>
:plan.initPlan <>
:plan.extParam (b 0)
:plan.allParam (b 0)
:hashkeys (
{VAR
:varno -2
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 3
:varattnosyn 1
:location 289
}
)
:skewTable 0
:skewColumn 0
:skewInherit false
:rows_total 2000306.4
}
:join.plan.initPlan <>
:join.plan.extParam (b 0)
:join.plan.allParam (b 0)
:join.jointype 4
:join.inner_unique false
:join.joinqual <>
:hashclauses (
{OPEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno -2
:varattno 2
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location 302
}
{VAR
:varno -1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 3
:varattnosyn 1
:location 289
}
}
)
:location -1
}
)
:hashoperators (o 96)
:hashcollations (o 0)
:hashkeys (
{VAR
:varno -2
:varattno 2
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location 302
}
)
}
:plan.righttree <>
:plan.initPlan <>
:plan.extParam (b 0)
:plan.allParam (b 0)
:aggstrategy 2
:aggssplit 6
:numCols 1
:grpCollIdx ( 1)
:grpOperators ( 98)
:grpCollations ( 100)
:numGroups 200
:transitionSpace 0
:aggParams (b)
:groupingSets <>
:chain <>
}
:plan.righttree <>
:plan.initPlan <>
:plan.extParam (b 0)
:plan.allParam (b 0)
:numCols 1
:sortCollIdx ( 1)
:sortOperators ( 664)
:collations ( 100)
:nullsFirst ( false)
}
:plan.righttree <>
:plan.initPlan <>
:plan.extParam (b)
:plan.allParam (b)
:num_workers 2

```

```

:rescan_param 0
:numCols 1
:sortColIdx ( 1)
:sortOperators ( 664)
:collations ( 100)
:nullsFirst ( false)
:initParam (b)
}
:plan.righttree <
:plan.initPlan <
:plan.extParam (b)
:plan.allParam (b)
:aggstrategy 1
:aggssplit 9
:numCols 1
:grpColIdx ( 1)
:grpOperators ( 98)
:grpCollations ( 100)
:numGroups 200
:transitionSpace 0
:aggParams (b)
:groupingSets <
:chain <
}

```

QUERY PLAN

```

Finalize GroupAggregate (cost=199632.93..199693.24 rows=200 width=17) (actual time=1444.081..1468.450 rows=5.00 loops=1)
Output: orders_secure.o_orderpriority, count(*)
Group Key: orders_secure.o_orderpriority
Buffers: shared hit=15197 read=119219, temp read=20117 written=20524
-> Gather Merge (cost=199632.93..199688.84 rows=480 width=17) (actual time=1444.069..1468.439 rows=15.00 loops=1)
Output: orders_secure.o_orderpriority, (PARTIAL count(*))
Workers Planned: 2
Workers Launched: 2
Buffers: shared hit=15197 read=119219, temp read=20117 written=20524
-> Sort (cost=198632.91..198633.41 rows=200 width=17) (actual time=1440.003..1440.010 rows=5.00 loops=3)
Output: orders_secure.o_orderpriority, (PARTIAL count(*))
Sort Key: orders_secure.o_orderpriority
Sort Method: quicksort Memory: 25kB
Buffers: shared hit=15197 read=119219, temp read=20117 written=20524
Worker 0: actual time=1438.109..1438.113 rows=5.00 loops=1
Sort Method: quicksort Memory: 25kB
Buffers: shared hit=5116 read=40207, temp read=6708 written=6876
Worker 1: actual time=1438.146..1438.158 rows=5.00 loops=1
Sort Method: quicksort Memory: 25kB
Buffers: shared hit=4746 read=40413, temp read=6640 written=6888
-> Partial HashAggregate (cost=198623.26..198625.26 rows=200 width=17) (actual time=1439.738..1439.745 rows=5.00 loops=3)
Output: orders_secure.o_orderpriority, PARTIAL count(*)
Group Key: orders_secure.o_orderpriority
Batches: 1 Memory Usage: 40kB
Buffers: shared hit=15181 read=119219, temp read=20117 written=20524
Worker 0: actual time=1437.804..1437.808 rows=5.00 loops=1
Batches: 1 Memory Usage: 40kB
Buffers: shared hit=5108 read=40207, temp read=6708 written=6876
Worker 1: actual time=1437.775..1437.780 rows=5.00 loops=1
Batches: 1 Memory Usage: 40kB
Buffers: shared hit=4738 read=40413, temp read=6640 written=6888
-> Parallel Hash Semi Join (cost=154455.47..198460.27 rows=32598 width=9) (actual time=1305.278..1438.355 rows=7772.33 loops=3)
Output: orders_secure.o_orderpriority
Hash Cond: (orders_secure.o_orderkey = lineitem.l_orderkey)
Buffers: shared hit=15181 read=119219, temp read=20117 written=20524
Worker 0: actual time=1300.444..1436.389 rows=7978.00 loops=1
Buffers: shared hit=5108 read=40207, temp read=6708 written=6876
Worker 1: actual time=1293.001..1436.360 rows=7787.00 loops=1
Buffers: shared hit=4738 read=40413, temp read=6640 written=6888
-> Subquery Scan on orders_secure (cost=4806.41..44383.43 rows=65196 width=13) (actual time=18.771..224.177 rows=8461.00 loops=3)
Output: orders_secure.o_orderpriority, orders_secure.o_orderkey
Filter: (orders_secure.o_orderdate < '1994-04-01 00:00:00'::timestamp without time zone)
Rows Removed by Filter: 149480
Buffers: shared hit=4647 read=25033
Worker 0: actual time=18.637..224.697 rows=8405.00 loops=1
Buffers: shared hit=1543 read=8247
Worker 1: actual time=19.019..223.987 rows=8538.00 loops=1
Buffers: shared hit=1567 read=8380
-> Parallel Hash Semi Join (cost=4806.41..41938.58 rows=195588 width=129) (actual time=18.536..213.502 rows=157941.33 loops=3)
Output: o.o_orderkey, NULL::integer, NULL::text, NULL::double precision, o.o_orderdate, o.o_orderpriority, NULL::text, NULL::integer, NULL::text
Hash Cond: (o.o_custkey = c.c_custkey)
Buffers: shared hit=4647 read=25033
Worker 0: actual time=18.613..214.220 rows=156322.00 loops=1
Buffers: shared hit=1543 read=8247
Worker 1: actual time=18.658..212.967 rows=158636.00 loops=1
Buffers: shared hit=1567 read=8380
-> Parallel Seq Scan on public.orders o (cost=0.00..33820.50 rows=432666 width=21) (actual time=1.190..101.082 rows=340755.33 loops=3)
Output: o.o_orderkey, o.o_custkey, o.o_orderstatus, o.o_totalprice, o.o_orderdate, o.o_orderpriority, o.o_clerk, o.o_shippriority, o.o_comment
Filter: (o.o_orderdate >= '1994-01-01'::date)
Rows Removed by Filter: 151245
Buffers: shared hit=975 read=25033
Worker 0: actual time=1.273..102.175 rows=344842.00 loops=1
Buffers: shared hit=312 read=8247
Worker 1: actual time=1.317..100.168 rows=350795.00 loops=1
Buffers: shared hit=348 read=8380
-> Parallel Hash (cost=4453.25..4453.25 rows=28253 width=4) (actual time=17.213..17.213 rows=22663.00 loops=3)
Output: c.c_custkey
Buckets: 131072 Batches: 1 Memory Usage: 3712kB
Buffers: shared hit=3672
Worker 0: actual time=17.046..17.046 rows=22784.00 loops=1
Buffers: shared hit=1231
Worker 1: actual time=17.288..17.288 rows=22538.00 loops=1
Buffers: shared hit=1219
-> Parallel Seq Scan on public.customer c (cost=0.00..4453.25 rows=28253 width=4) (actual time=0.026..13.275 rows=22663.00 loops=3)
Output: c.c_custkey
Filter: (c.c_actbal > '5000'::double precision)
Rows Removed by Filter: 27337
Buffers: shared hit=3672
Worker 0: actual time=0.025..13.324 rows=22784.00 loops=1
Buffers: shared hit=1231
Worker 1: actual time=0.029..13.253 rows=22538.00 loops=1
Buffers: shared hit=1219
-> Parallel Hash (cost=135974.79..135974.79 rows=833461 width=4) (actual time=1056.095..1056.095 rows=1264432.00 loops=3)
Output: lineitem.l_orderkey
Buckets: 262144 (originally 262144) Batches: 32 (originally 16) Memory Usage: 6784kB
Buffers: shared hit=10534 read=94186, temp read=9108 written=20140
Worker 0: actual time=1054.494..1054.494 rows=1286355.00 loops=1
Buffers: shared hit=3565 read=31960, temp read=3014 written=6748
Worker 1: actual time=1054.490..1054.490 rows=1275598.00 loops=1
Buffers: shared hit=3171 read=32033, temp read=3055 written=6760
-> Parallel Seq Scan on public.lineitem (cost=0.00..135974.79 rows=833461 width=4) (actual time=0.617..553.293 rows=1264432.00 loops=3)
Output: lineitem.l_orderkey
Filter: (lineitem.l_commitdate < lineitem.l_receiptdate)
Rows Removed by Filter: 735973
Buffers: shared hit=18534 read=94186
Worker 0: actual time=0.340..562.332 rows=1286355.00 loops=1
Buffers: shared hit=3565 read=31960
Worker 1: actual time=0.652..550.597 rows=1275598.00 loops=1
Buffers: shared hit=3171 read=32033

```

```

Planning:
Buffers: shared hit=209
Planning Time: 7.507 ms
Execution Time: 1469.436 ms
(103 rows)

```

tpch=>